

Module 1: Basic Concepts

1. HelloWorld.java

```
public class HelloWorld {  
    public static void main(String[] args) {  
        System.out.println("Hello, Java World!");  
    }  
}
```

2. MyInfo.java

```
public class MyInfo {  
    public static void main(String[] args) {  
        int age = 21;  
        double height = 1.75;  
        char initial = 'A';  
        boolean isStudent = true;  
        String fullName = "Your Name";  
        System.out.println("Age: " + age);  
        System.out.println("Height: " + height);  
        System.out.println("Initial: " + initial);  
        System.out.println("Is Student: " + isStudent);  
        System.out.println("Full Name: " + fullName);  
    }  
}
```

3. ArithmeticDemo.java

```
public class ArithmeticDemo {  
    public static void main(String[] args) {  
        double num1 = 25.5, num2 = 7.3;  
        System.out.println("Addition: " + (num1 + num2));  
        System.out.println("Subtraction: " + (num1 - num2));  
        System.out.println("Multiplication: " + (num1 * num2));  
        System.out.println("Division: " + (num1 / num2));  
        System.out.println("Modulus: " + (num1 % num2));  
    }  
}
```

```
}
```

4. SumCalculator.java

```
import java.util.Scanner;
public class SumCalculator {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter two integers: ");
        int a = sc.nextInt(), b = sc.nextInt();
        System.out.println("The sum of " + a + " and " + b + " is " + (a + b));
    }
}
```

5. RectangleArea.java

```
import java.util.Scanner;
public class RectangleArea {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter length and width: ");
        double l = sc.nextDouble(), w = sc.nextDouble();
        System.out.println("Area: " + (l * w));
    }
}
```

6. RelationalOps.java

```
public class RelationalOps {
    public static void main(String[] args) {
        int val1 = 50, val2 = 30;
        System.out.println("val1 > val2: " + (val1 > val2));
        System.out.println("val1 < val2: " + (val1 < val2));
        System.out.println("val1 == val2: " + (val1 == val2));
        System.out.println("val1 != val2: " + (val1 != val2));
    }
}
```

7. LogicalOps.java

```
public class LogicalOps {
    public static void main(String[] args) {
        boolean isRaining = true, hasUmbrella = false, isCold = true;
        System.out.println("AND: " + (isRaining && hasUmbrella));
    }
}
```

```

        System.out.println("OR: " + (isRaining || hasUmbrella));
        System.out.println("NOT: " + (!isRaining));
        System.out.println("Complex: " + (isRaining && (hasUmbrella || isCold)));
    }
}

```

8. TempConverter.java

```

import java.util.Scanner;
public class TempConverter {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter Celsius: ");
        double c = sc.nextDouble();
        System.out.println("Fahrenheit: " + (c * 9.0/5.0 + 32.0));
    }
}

```

9. AssignmentOps.java

```

public class AssignmentOps {
    public static void main(String[] args) {
        int x = 10;
        x += 5; System.out.println("+=: " + x);
        x = 10; x -= 3; System.out.println("-=: " + x);
        x = 10; x *= 2; System.out.println("*=: " + x);
        x = 10; x /= 4; System.out.println("/=: " + x);
    }
}

```

10. IncDecOps.java

```

public class IncDecOps {
    public static void main(String[] args) {
        int a = 5;
        System.out.println("Post-inc: " + (a++)); // 5
        System.out.println("Result: " + a);    // 6
        a = 5;
        System.out.println("Pre-inc: " + (++a)); // 6
    }
}

```

Module 2: Control Structures

11.NumberClassifier.java

```
import java.util.Scanner;
public class NumberClassifier {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter an integer: ");
        int num = sc.nextInt();

        if (num > 0) {
            System.out.println("The number is positive.");
        } else if (num < 0) {
            System.out.println("The number is negative.");
        } else {
            System.out.println("The number is zero.");
        }
        sc.close();
    }
}
```

12. VoterEligibility.java

```
import java.util.Scanner;
public class VoterEligibility {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter your age: ");
        int age = sc.nextInt();

        if (age >= 18) {
            System.out.println("You are eligible to vote.");
        } else {
            System.out.println("You are not yet eligible to vote.");
        }
        sc.close();
    }
}
```

13.LargestNumber.java

```

import java.util.Scanner;
public class LargestNumber {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter three integers: ");
        int n1 = sc.nextInt();
        int n2 = sc.nextInt();
        int n3 = sc.nextInt();

        if (n1 >= n2 && n1 >= n3) {
            System.out.println(n1 + " is the largest.");
        } else if (n2 >= n1 && n2 >= n3) {
            System.out.println(n2 + " is the largest.");
        } else {
            System.out.println(n3 + " is the largest.");
        }
        sc.close();
    }
}

```

14. GradeCalculator.java

```

import java.util.Scanner;
public class GradeCalculator {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter score (0-100): ");
        int score = sc.nextInt();

        if (score < 0 || score > 100) {
            System.out.println("Error: Invalid score.");
        } else if (score >= 90) {
            System.out.println("Grade: A");
        } else if (score >= 80) {
            System.out.println("Grade: B");
        } else if (score >= 70) {
            System.out.println("Grade: C");
        } else if (score >= 60) {
            System.out.println("Grade: D");
        } else {
            System.out.println("Grade: F");
        }
        sc.close();
    }
}

```

```
}  
}
```

15. DayOfWeek.java

```
import java.util.Scanner;  
public class DayOfWeek {  
    public static void main(String[] args) {  
        Scanner sc = new Scanner(System.in);  
        System.out.print("Enter a number (1-7): ");  
        int dayNum = sc.nextInt();  
  
        switch (dayNum) {  
            case 1: System.out.println("Monday"); break;  
            case 2: System.out.println("Tuesday"); break;  
            case 3: System.out.println("Wednesday"); break;  
            case 4: System.out.println("Thursday"); break;  
            case 5: System.out.println("Friday"); break;  
            case 6: System.out.println("Saturday"); break;  
            case 7: System.out.println("Sunday"); break;  
            default: System.out.println("Invalid day.");  
        }  
        sc.close();  
    }  
}
```

16. MenuProgram.java

```
import java.util.Scanner;  
public class MenuProgram {  
    public static void main(String[] args) {  
        Scanner sc = new Scanner(System.in);  
        System.out.println("Menu:");  
        System.out.println("1. Option A");  
        System.out.println("2. Option B");  
        System.out.println("3. Exit");  
        System.out.print("Enter choice: ");  
        int choice = sc.nextInt();  
  
        switch (choice) {  
            case 1:  
                System.out.println("You chose Option A.");  
                break;  
            case 2:  
                System.out.println("You chose Option B.");  
                break;  
            case 3:  
                System.out.println("Exiting...");  
                break;  
            default:  
                System.out.println("Invalid choice.");  
                break;  
        }  
    }  
}
```

```

        System.out.println("You chose Option B.");
        break;
    case 3:
        System.out.println("Exiting program. Goodbye!");
        break;
    default:
        System.out.println("Invalid choice. Please try again.");
    }
    sc.close();
}
}

```

17. ForLoopDemo.java

```

public class ForLoopDemo {
    public static void main(String[] args) {
        for(int i = 1; i <= 10; i++) System.out.println(i);
    }
}

```

18. SumNNumbers.java

```

public class SumNNumbers {
    public static void main(String[] args) {
        int n = 10, sum = 0;
        for(int i = 1; i <= n; i++) sum += i;
        System.out.println("Sum: " + sum);
    }
}

```

19. MultiplicationTable.java

```

public class MultiplicationTable {
    public static void main(String[] args) {
        int n = 5;
        for(int i = 1; i <= 10; i++) System.out.println(n + " x " + i + " = " + (n*i));
    }
}

```

20. WhileLoopCountdown.java

```

public class WhileLoopCountdown {
    public static void main(String[] args) {
        int i = 10;
    }
}

```

```

        while(i >= 1) { System.out.println(i); i--; }
    }
}

```

21. Factorial.java

```

public class Factorial {
    public static void main(String[] args) {
        int n = 5, fact = 1;
        while(n > 0) { fact *= n; n--; }
        System.out.println("Factorial: " + fact);
    }
}

```

22. SumOfDigits.java

```

public class SumOfDigits {
    public static void main(String[] args) {
        int n = 123, sum = 0;
        while(n > 0) { sum += n % 10; n /= 10; }
        System.out.println("Sum: " + sum);
    }
}

```

23.PasswordPrompt.java

```

import java.util.Scanner;
public class PasswordPrompt {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        String predefinedPassword = "java123";
        String userInput;

        // The do-while loop ensures the prompt appears at least once
        do {
            System.out.print("Enter the password: ");
            userInput = sc.nextLine();

            if (!userInput.equals(predefinedPassword)) {
                System.out.println("Incorrect password. Try again.");
            }
        } while (!userInput.equals(predefinedPassword));

        System.out.println("Access granted!");
    }
}

```



```

        sc.close();
    }
}

```

24. InteractiveMenu.java

```

import java.util.Scanner;
public class InteractiveMenu {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int choice;

        do {
            System.out.println("\n--- Interactive Menu ---");
            System.out.println("1. Add");
            System.out.println("2. Subtract");
            System.out.println("3. Exit");
            System.out.print("Enter your choice: ");
            choice = sc.nextInt();

            switch (choice) {
                case 1:
                    System.out.print("Enter two numbers to add: ");
                    double a1 = sc.nextDouble();
                    double b1 = sc.nextDouble();
                    System.out.println("Result: " + (a1 + b1));
                    break;
                case 2:
                    System.out.print("Enter two numbers to subtract: ");
                    double a2 = sc.nextDouble();
                    double b2 = sc.nextDouble();
                    System.out.println("Result: " + (a2 - b2));
                    break;
                case 3:
                    System.out.println("Goodbye!");
                    break;
                default:
                    System.out.println("Invalid choice. Please try again.");
            }
        } while (choice != 3);

        sc.close();
    }
}

```

Assignment 25. EvenOddCounter.java

```
import java.util.Scanner;
public class EvenOddCounter {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter a positive integer N: ");
        int n = sc.nextInt();

        int evenCount = 0;
        int oddCount = 0;

        for (int i = 1; i <= n; i++) {
            if (i % 2 != 0) {
                oddCount++;
                continue; // Skips the rest of the loop for odd numbers
            }
            // This part only runs if the number is even
            evenCount++;
        }

        System.out.println("In the range 1 to " + n + ":");
        System.out.println("Even numbers: " + evenCount);
        System.out.println("Odd numbers: " + oddCount);
        sc.close();
    }
}
```

26.PrimeChecker.java

```
import java.util.Scanner;
public class PrimeChecker {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter an integer greater than 1: ");
        int num = sc.nextInt();

        boolean isPrime = true;

        if (num <= 1) {
            isPrime = false;
        } else {
            // Check for divisors from 2 up to the square root of num
            for (int i = 2; i <= Math.sqrt(num); i++) {
                if (num % i == 0) {
```

```

        isPrime = false;
        break; // Exit loop immediately as it's not a prime
    }
}

if (isPrime) {
    System.out.println(num + " is a prime number.");
} else {
    System.out.println(num + " is not a prime number.");
}
sc.close();
}
}

```

27.ReverseNumber.java

```

import java.util.Scanner;

public class ReverseNumber {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter a positive integer: ");
        int num = sc.nextInt();
        int originalNum = num;
        int reversedNum = 0;

        while (num > 0) {
            int digit = num % 10;    // Extract the last digit
            reversedNum = reversedNum * 10 + digit; // Build the reversed number
            num /= 10;               // Remove the last digit
        }

        System.out.println("Original Number: " + originalNum);
        System.out.println("Reversed Number: " + reversedNum);
        sc.close();
    }
}

```

28. Fibonacci.java

```

import java.util.Scanner;

public class Fibonacci {
    public static void main(String[] args) {

```

```

Scanner sc = new Scanner(System.in);
System.out.print("Enter the number of terms (N): ");
int n = sc.nextInt();

int firstTerm = 0, secondTerm = 1;
System.out.print("Fibonacci Sequence: ");

for (int i = 1; i <= n; i++) {
    System.out.print(firstTerm + " ");

    // Compute the next term
    int nextTerm = firstTerm + secondTerm;
    firstTerm = secondTerm;
    secondTerm = nextTerm;
}
sc.close();
}
}

```

29. Pattern1.java

```

public class Pattern1 {
    public static void main(String[] args) {
        int rows = 5;

        for (int i = 1; i <= rows; i++) {    // Outer loop for rows
            for (int j = 1; j <= i; j++) {    // Inner loop for columns
                System.out.print("*");
            }
            System.out.println();           // Move to the next line
        }
    }
}

```

30. Pattern2.java

```

public class Pattern2 {
    public static void main(String[] args) {
        int rows = 5;

        for (int i = 1; i <= rows; i++) {    // Outer loop for rows
            for (int j = 1; j <= i; j++) {    // Inner loop for printing numbers
                System.out.print(j);
            }
            System.out.println();           // Move to the next line
        }
    }
}

```

```
    }  
  }  
}
```

31: Person.java

```
public class Person {  
    public String name;  
    public int age;  
  
    public void introduce() {  
        System.out.println("Hi, my name is " + name + " and I am " + age + " years old.");  
    }  
  
    public static void main(String[] args) {  
        Person p1 = new Person();  
        p1.name = "Alice";  
        p1.age = 25;  
        p1.introduce();  
  
        Person p2 = new Person();  
        p2.name = "Bob";  
        p2.age = 30;  
        p2.introduce();  
    }  
}
```

32: Book.java

```
public class Book {  
    private String title;  
    private String author;  
    private double price;  
  
    // Getters  
    public String getTitle() { return title; }  
    public String getAuthor() { return author; }  
    public double getPrice() { return price; }  
  
    // Setter for price  
    public void setPrice(double newPrice) {  
        if (newPrice >= 0) {  
            this.price = newPrice;  
        }  
    }  
}
```

```

public static void main(String[] args) {
    Book myBook = new Book();
    myBook.title = "Java Programming"; // Direct access allowed if in same file/package logic
    myBook.author = "John Smith";
    myBook.setPrice(29.99);

    System.out.println("Book: " + myBook.getTitle() + " | Price: $" + myBook.getPrice());
}
}

```

33: Product.java

```

public class Product {
    private String productId;
    private String name;
    private int stock;

    public Product(String id, String name, int stock) {
        this.productId = id;
        this.name = name;
        this.stock = stock;
    }

    public void displayProductInfo() {
        System.out.println("ID: " + productId + ", Name: " + name + ", Stock: " + stock);
    }

    public static void main(String[] args) {
        Product p1 = new Product("P101", "Laptop", 10);
        Product p2 = new Product("P102", "Mouse", 50);
        p1.displayProductInfo();
        p2.displayProductInfo();
    }
}

```

Assignment 34: Circle.java

```

public class Circle {
    private double radius;

    public Circle(double radius) {
        this.radius = radius;
    }
}

```

```

    public double getRadius() { return radius; }

    public double calculateArea() {
        return Math.PI * Math.pow(radius, 2);
    }

    public static void main(String[] args) {
        Circle c = new Circle(5.0);
        System.out.println("Radius: " + c.getRadius());
        System.out.println("Area: " + c.calculateArea());
    }
}

```

35: Student.java

```

public class Student {
    private int studentId;
    private String name;
    private String major;

    // Default Constructor
    public Student() {
        this.studentId = 0;
        this.name = "Unknown";
        this.major = "Undecided";
    }

    // Parameterized Constructor
    public Student(int id, String name, String major) {
        this.studentId = id;
        this.name = name;
        this.major = major;
    }

    public void displayStudentDetails() {
        System.out.println("ID: " + studentId + ", Name: " + name + ", Major: " + major);
    }

    public static void main(String[] args) {
        Student s1 = new Student();
        Student s2 = new Student(101, "Kiran", "Computer Science");
        s1.displayStudentDetails();
        s2.displayStudentDetails();
    }
}

```

36: BankAccount.java

```
public class BankAccount {
    private String accountNumber;
    private String accountHolderName;
    private double balance;

    public BankAccount(String accNum, String name) {
        this.accountNumber = accNum;
        this.accountHolderName = name;
        this.balance = 0.0;
    }

    public void deposit(double amount) {
        if (amount > 0) {
            balance += amount;
            System.out.println("Deposited: " + amount + ". New Balance: " + balance);
        } else {
            System.out.println("Invalid deposit amount.");
        }
    }

    public void withdraw(double amount) {
        if (amount > 0 && amount <= balance) {
            balance -= amount;
            System.out.println("Withdrew: " + amount + ". New Balance: " + balance);
        } else {
            System.out.println("Withdrawal failed: Insufficient funds or invalid amount.");
        }
    }

    public double getBalance() { return balance; }

    public static void main(String[] args) {
        BankAccount acc = new BankAccount("SBI123", "Arjun");
        acc.deposit(500);
        acc.withdraw(200);
        acc.withdraw(400); // Should fail
    }
}
```

37: Dog.java

```
public class Dog {
```



```

private String name;
private String breed;

public Dog(String name, String breed) {
    this.name = name;
    this.breed = breed;
}

public void bark() {
    System.out.println(name + " says Woof!");
}

public void eat() {
    System.out.println(name + " the " + breed + " is eating.");
}

public static void main(String[] args) {
    Dog d1 = new Dog("Buddy", "Golden Retriever");
    d1.bark();
    d1.eat();
}
}

```

38: Employee.java

```

public class Employee {
    private int employeeId;
    private String name;
    private double hourlyRate;
    private int hoursWorked;

    public Employee(int id, String name, double rate, int hours) {
        this.employeeId = id;
        this.name = name;
        this.hourlyRate = rate;
        this.hoursWorked = hours;
    }

    public double calculateMonthlySalary() {
        return hourlyRate * hoursWorked;
    }

    public static void main(String[] args) {
        Employee emp = new Employee(501, "Rahul", 25.0, 160);
        System.out.println("Employee: " + emp.name);
    }
}

```

```

        System.out.println("Monthly Salary: $" + emp.calculateMonthlySalary());
    }
}

```

39: InventoryItem.java

```

public class InventoryItem {
    private String itemName;
    private int quantityInStock;

    public InventoryItem(String name, int quantity) {
        this.itemName = name;
        this.quantityInStock = quantity;
    }

    public void restock(int quantity) {
        if (quantity > 0) quantityInStock += quantity;
    }

    public void sell(int quantity) {
        if (quantity > 0 && quantity <= quantityInStock) {
            quantityInStock -= quantity;
        } else {
            System.out.println("Out of stock!");
        }
    }

    public int getStock() { return quantityInStock; }

    public static void main(String[] args) {
        InventoryItem item = new InventoryItem("Pen", 100);
        item.sell(30);
        System.out.println("Remaining Stock: " + item.getStock());
    }
}

```

40: TemperatureConverter.java

```

public class TemperatureConverter {

    public static double celsiusToFahrenheit(double celsius) {
        return (celsius * 9.0 / 5.0) + 32.0;
    }

    public static double fahrenheitToCelsius(double fahrenheit) {

```

```

        return (fahrenheit - 32.0) * 5.0 / 9.0;
    }

    public static void main(String[] args) {
        double c = 37.0;
        System.out.println(c + "°C is " + TemperatureConverter.celsiusToFahrenheit(c) + "°F");

        double f = 98.6;
        System.out.println(f + "°F is " + TemperatureConverter.fahrenheitToCelsius(f) + "°C");
    }
}

```

41: ShapeRectangle.java

```

class Shape {
    public void displayInfo() {
        System.out.println("This is a generic shape.");
    }
}

class Rectangle extends Shape {
    private double length, width;
    public Rectangle(double l, double w) {
        length = l; width = w;
    }
}

public class ShapeRectangle {
    public static void main(String[] args) {
        Rectangle rect = new Rectangle(10, 5);
        rect.displayInfo(); // Inherited method
    }
}

```

42: AnimalCat.java

```

class Animal {
    public void makeSound() {
        System.out.println("Animal makes a sound.");
    }
}

class Cat extends Animal {
    @Override
    public void makeSound() {

```

```

        System.out.println("Cat meows.");
    }
}

public class AnimalCat {
    public static void main(String[] args) {
        Animal myAnimal = new Animal();
        Cat myCat = new Cat();
        myAnimal.makeSound();
        myCat.makeSound(); // Overridden method
    }
}

```

43: VehicleCar.java

```

class Vehicle {
    protected String brand;
    protected int year;

    public Vehicle(String brand, int year) {
        this.brand = brand;
        this.year = year;
    }
}

class Car extends Vehicle {
    private int numberOfDoors;

    public Car(String brand, int year, int doors) {
        super(brand, year); // Calling parent constructor
        this.numberOfDoors = doors;
    }

    public void displayInfo() {
        System.out.println("Brand: " + brand + ", Year: " + year + ", Doors: " + numberOfDoors);
    }
}

public class VehicleCar {
    public static void main(String[] args) {
        Car myCar = new Car("Toyota", 2024, 4);
        myCar.displayInfo();
    }
}

```

44: AreaCalculator.java

```
public class AreaCalculator {  
    public static double calculateArea(double side) {  
        return side * side;  
    }  
  
    public static double calculateArea(double length, double width) {  
        return length * width;  
    }  
  
    public static void main(String[] args) {  
        System.out.println("Square Area: " + calculateArea(5.0));  
        System.out.println("Rectangle Area: " + calculateArea(10.0, 5.0));  
    }  
}
```

45: RuntimePolymorphism.java

```
class Employee {  
    public void calculateSalary() {  
        System.out.println("Calculating generic employee salary.");  
    }  
}  
  
class FullTimeEmployee extends Employee {  
    @Override  
    public void calculateSalary() {  
        System.out.println("Calculating full-time employee salary (Fixed).");  
    }  
}  
  
public class RuntimePolymorphism {  
    public static void main(String[] args) {  
        Employee emp = new FullTimeEmployee(); // Upcasting  
        emp.calculateSalary(); // Calls child class method at runtime  
    }  
}
```

46: AbstractShape.java

```
abstract class Shape {  
    protected String name;  
    public Shape(String name) { this.name = name; }  
    public abstract double calculateArea(); // No implementation  
}
```

```

class Circle extends Shape {
    private double radius;
    public Circle(double r) { super("Circle"); this.radius = r; }

    public double calculateArea() { return Math.PI * radius * radius; }
}

```

```

public class AbstractShape {
    public static void main(String[] args) {
        Circle c = new Circle(5.0);
        System.out.println("Area: " + c.calculateArea());
    }
}

```

47: AbstractVehicle.java

```

abstract class Vehicle {
    public void startEngine() { System.out.println("Engine started."); }
    public abstract void drive();
}

```

```

class Bike extends Vehicle {
    public void drive() { System.out.println("Riding the bike."); }
}

```

```

public class AbstractVehicle {
    public static void main(String[] args) {
        Bike myBike = new Bike();
        myBike.startEngine();
        myBike.drive();
    }
}

```

48: WalkableInterface.java

```

interface Walkable {
    void walk();
}

```

```

class Human implements Walkable {
    public void walk() { System.out.println("Human is walking."); }
}

```

```

class Dog implements Walkable {

```

```
    public void walk() { System.out.println("Dog is walking."); }  
}
```

```
public class WalkableInterface {  
    public static void main(String[] args) {  
        Walkable h = new Human();  
        Walkable d = new Dog();  
        h.walk();  
        d.walk();  
    }  
}
```

49: PlayableInterface.java

```
interface Playable {  
    void play();  
}
```

```
class Guitar implements Playable {  
    public void play() { System.out.println("Strumming the guitar."); }  
}
```

```
class Piano implements Playable {  
    public void play() { System.out.println("Playing the piano keys."); }  
}
```

```
public class PlayableInterface {  
    public static void main(String[] args) {  
        Playable g = new Guitar();  
        Playable p = new Piano();  
        g.play();  
        p.play();  
    }  
}
```

50: MediaScenario.java

```
abstract class MediaItem {  
  
    protected String title;  
    public abstract void play();  
}
```

```
interface Sharable {  
    void share();  
}
```

```
}
```

```
class Movie extends MediaItem implements Sharable {  
    public Movie(String t) { this.title = t; }  
    public void play() { System.out.println("Playing Movie: " + title); }  
    public void share() { System.out.println("Sharing Movie link..."); }  
}
```

```
public class MediaScenario {  
    public static void main(String[] args) {  
        Movie m = new Movie("Inception");  
        m.play();  
        m.share();  
    }  
}
```

51. PalindromeChecker.java

```
public class PalindromeChecker {  
    public static void main(String[] args) {  
        String original = "madam", reverse = "";  
        for (int i = original.length() - 1; i >= 0; i--) reverse += original.charAt(i);  
        System.out.println(original.equals(reverse) ? "Palindrome" : "Not a Palindrome");  
    }  
}
```

52. ArrayDemo.java

```
public class ArrayDemo {  
    public static void main(String[] args) {  
        int[] numbers = {10, 20, 30, 40, 50};  
        int sum = 0;  
        for (int n : numbers) {  
            System.out.println(n);  
            sum += n;  
        }  
        System.out.println("Sum: " + sum);  
    }  
}
```

53. ArrayMax.java

```
public class ArrayMax {  
    public static void main(String[] args) {
```



```

        int[] arr = {10, 5, 20, 15, 8};
        int max = arr[0];
        for (int i = 1; i < arr.length; i++) if (arr[i] > max) max = arr[i];
        System.out.println("Max: " + max);
    }
}

```

54. Calculator.java

```

class Calculator {
    public static int add(int a, int b) { return a + b; }
    public static int subtract(int a, int b) { return a - b; }
}

public class Main {
    public static void main(String[] args) {
        System.out.println("Sum: " + Calculator.add(10, 20));
    }
}

```

55. PolymorphicShapesArray.java

```

public class PolymorphicShapesArray {
    public static void main(String[] args) {
        Shape[] shapes = new Shape[2];
        shapes[0] = new Circle(5.0);
        shapes[1] = new Circle(10.0); // Or Rectangle if implemented
        for (Shape s : shapes) System.out.println("Area: " + s.calculateArea());
    }
}

```

51. PalindromeChecker.java

```

public class PalindromeChecker {
    public static void main(String[] args) {
        String s = "madam", rev = "";
        for(int i = s.length()-1; i >= 0; i--) rev += s.charAt(i);
        System.out.println(s.equals(rev) ? "Palindrome" : "Not");
    }
}

```

52. ArrayDemo.java

```

public class ArrayDemo {
    public static void main(String[] args) {
        int[] numbers = {10, 20, 30, 40, 50};
        int sum = 0;
        for(int n : numbers) { System.out.println(n); sum += n; }
        System.out.println("Total: " + sum);
    }
}

```

55. PolymorphicShapesArray.java

```

abstract class Shape { abstract double area(); }
class Circle extends Shape {
    double r; Circle(double r) { this.r = r; }
    double area() { return Math.PI * r * r; }
}
class Rectangle extends Shape {
    double l, w; Rectangle(double l, double w) { this.l = l; this.w = w; }
    double area() { return l * w; }
}

public class Main {
    public static void main(String[] args) {
        Shape[] shapes = { new Circle(5), new Rectangle(10, 2) };
        for(Shape s : shapes) System.out.println("Area: " + s.area());
    }
}

```